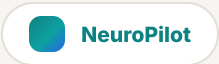


System Blueprint

End-to-end architecture — data → training pipeline → model artifacts → serving → decision API → dashboard → persistence. One command runs the whole loop.



1 · DATA SOURCES

INPUT

Real OASIS-1 longitudinal
150 subjects · 373 visits · SES/MMSE gaps audited, never silently dropped

INPUT

Synthetic ADNI-shaped cohort v2
800 subjects · 200 CN / 400 MCI / 200 AD · all 4 stages measured

INPUT

12-month follow-up simulation
biomarker-driven drift · 133 conversions · published conversion base rates



2 · TRAINING PIPELINE (PYTHON SCRIPTS/)

ETL

scripts/ingest.py
schema check · dedupe · missing-value audit · patients + visits tables

TRAIN

scripts/train_model.py
XGBoost pipeline + RF fallback · subject-level split · leakage guards

TRAIN

scripts/train_progression_model.py
MMSE-Δ regressor + conversion classifier · same 10-feature vector



3 · MODEL ARTIFACTS

ARTIFACTS

artifacts/ — git-tracked for deployment
pipeline.joblib · rf_pipeline.joblib · progression_delta.joblib · progression_conversion.joblib · model_meta.json · eval reports + global importance



4 · SERVING LAYER (BACKEND/APP)

SERVING

FastAPI app — backend/app
lifespan loads artifacts once · model_service + progression modules · SHAP TreeExplainer cached

SERVING

Escalation rule engine
transparent (stage × tier) → next-test matrix · clinician-in-the-loop · explicit override path

SERVING

Storage layer
in-memory cohort + optional Postgres mirror · 24 pytest cases cover the API surface



5 · DECISION API

API

Decision API — 14 typed endpoints
/patients · /patients/{id} · /explain · /pipeline · /progression · /compare · /score · /workup/* · /advance-stage · /results · /health · /model/info

API

OpenAPI + Swagger at /docs
every request/response typed with Pydantic · no diagnosis field anywhere — risk tier + reasoning only



6 · REACT DASHBOARD (SRC/)

UI

React 19 + Vite + Tailwind
Dashboard · Worklist · Patient record · Progression view · Priority Comparison · Risk Simulator

UI

Decision aids on the record
risk gauge with 0.4 / 0.7 threshold marks · SHAP attribution · event & reasoning trail · 12-mo forecast

UI

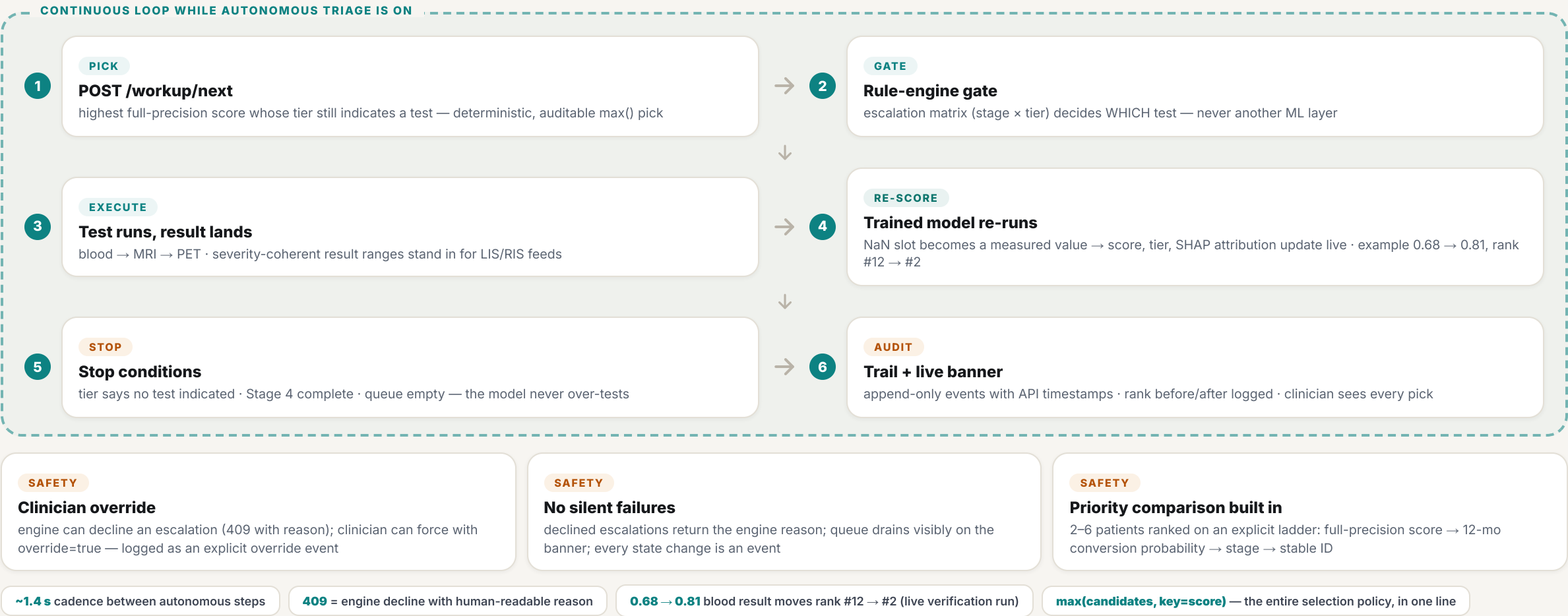
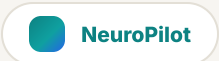
Autonomous Triage toggle
workup/next every ~1.4 s · the model picks patient AND test · every pick streams to the live banner



7 · PERSISTENCE & PACKAGING

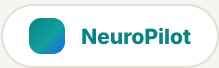
Autonomous Triage Loop

The runtime decision cycle: the model picks the patient AND the test, every result re-scores the cohort, and every step lands in the audit trail. Cognition drives blood; the blood-informed score drives MRI; and so on.



Model Layer

Two XGBoost model families on one shared 10-feature stage-aware vector. Missing tests are never imputed or invented — missingness is a learned, informative state; the pipeline orders the right test to resolve it.



SHARED FEATURE SPACE

SHARED INPUT

The 10-feature stage-aware vector

age · sex · education_years · mmse · mmse_change · ptau181 · abeta4240 · hippocampal_volume · amyloid_positive · tau_positive

Un-ordered tests stay **NaN** — XGBoost routes them down a learned default path; SHAP badges it `model default`. Measured-but-gappy values (OASIS SES/MMSE) are median-imputed inside the bundled sklearn pipeline — identical at train and serve time.



MODEL 1 · RISK SCORING

XGBoost classifier — who needs attention now

Severity-correlated labels · CDR excluded (label-leakage guard) · subject-level split · preprocessing bundled in the served artifact. Serves score, tier (0.4 / 0.7 thresholds), per-patient SHAP attribution, global importance and Risk Simulator re-scoring.

CV AUC

0.842

TEST AUC

0.871

ACCURACY

0.750

RF FALLBACK

0.874

MODEL 2 · PROGRESSION FORECAST

XGBoost pair — where the patient is headed

Same 10 features, two heads: MMSE-Δ regressor (expected 12-month point change ± uncertainty band) + conversion classifier (probability of crossing into the next phase: CN→MCI, MCI→AD, AD→severe). Projected tier = the risk model re-scoring the future vector.

Δ MAE

0.611

Δ R²

0.679

CONV. AUC

0.770

PR LIFT

2.8×



ONE EXPLAINABILITY SURFACE

EXPLAINABILITY

SHAP everywhere — same explainer family across both models

Per-patient waterfall attribution with measured values · global mean-|SHAP| importance chart · missing-stage `model default` badge (dimmed, footnoted) · progression forecast drivers · Risk Simulator live waterfall — a clinician can see WHY before trusting WHAT.

Guardrail · CDR excluded — clinician rating ≈ diagnosis (label leakage)

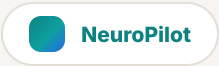
Guardrail · subject-level split — no subject in train AND test

Guardrail · imputation inside artifact — train/serve parity

Guardrail · no diagnosis field in any API contract

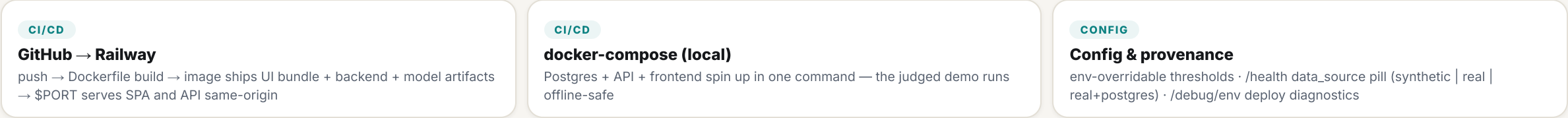
Guardrail · disclaimer on every forecast

Deployment & Trust Architecture

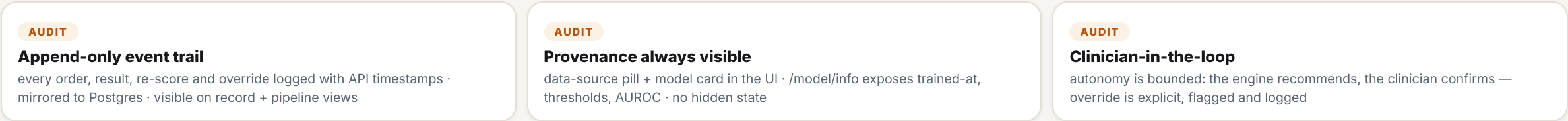


One container to production, Postgres-backed persistence, and the provenance rails that make every number on screen defensible under questioning.

1 · DEPLOYMENT TOPOLOGY



2 · TRUST & AUDIT RAILS



- Railway verified · real+postgres live · 800 subjects seeded · progression + compare endpoints 200 OK

24 pytest cases · API surface covered on every push

0 mock data · the dashboard reads only the decision API
- Artifacts in git · deploy contains the exact trained models